

1605

Agriculture and Crops of India

About 1 question a paper from the Indian Geography slot, plus spillover into Indian Economy and Current Affairs through MSP, PM-KISAN and crop insurance. Seasons, soils and the revolutions are the three blocks that repeat.

Agriculture supports nearly half of India's workforce but contributes only about 18 per cent of gross value added - that gap is the whole story of Indian farming. For the exam, though, this chapter is a matching chapter: crop to season, crop to state, soil to region, revolution to product, scheme to year.

Agriculture and Crops of India

Seasons

- Kharif - monsoon sown
- Rabi - winter sown
- Zaid - short summer

Plantation

- Tea-Assam
- Coffee-Karnataka
- Rubber-Kerala

Price and market

- CACP
- MSP 22 crops
- FRP sugarcane
- FCI
- e-NAM

Cereals

- Rice
- Wheat
- Bajra
- Jowar
- Maize

Soils

- Alluvial
- Black/regur
- Red-yellow
- Laterite
- Arid
- Saline

Schemes

- PM-KISAN
- PMFBY
- Soil Health Card
- NFSM
- RKVY
- PKVY

Cash crops

- Sugarcane
- Cotton
- Jute
- Oilseeds

Revolutions

- Green-foodgrain
- White-milk
- Blue-fish
- Yellow-oilseeds

The three cropping seasons

India's farming calendar is set by the monsoon, not by the English calendar. Kharif is sown when the south-west monsoon arrives and harvested when it leaves. Rabi is sown into the cooling soil after the monsoon and harvested in spring. Zaid is the short gap crop between them. Almost every year some paper somewhere asks you to pick the odd crop out of a season list, so learn the crop lists cold.

Kharif vs Rabi

Kharif	Rabi
Sown June-July, with the onset of the south-west monsoon	Sown October-December, after the monsoon withdraws
Harvested September-October	Harvested April-June
Needs high temperature and heavy rain	Needs a cool growing season and bright sunshine at ripening
Rice, jowar, bajra, maize, cotton, jute, groundnut, tur, soybean	Wheat, barley, gram, mustard, peas, linseed
Rain-fed over much of the country	Depends heavily on irrigation and winter western disturbances

- Zaid runs in the short season between the rabi harvest and the kharif sowing.
- Zaid crops: watermelon, muskmelon, cucumber, summer moong and fodder crops.
- Gram and mustard are rabi, not zaid - a standard planted error in statement questions.
- Barley is rabi: it is planted inside kharif lists as the odd one out in 'which is NOT a kharif crop'.

Foodgrains - conditions and leading states

For each major crop the examiner wants two things: the growing condition (rainfall and temperature) and the leading producer state. Do not chase the second and third ranks - those move with the year. Fix the leader, and know which leaders are stable.

- Rice - kharif, over 100 cm rainfall, about 25 degrees C; grown with canal and tube-well water in Punjab and Haryana where rainfall is low.
- West Bengal grows three rice crops - aus, aman and boro.
- Punjab has the highest rice yield per hectare among the major states.
- Wheat - rabi, 50-75 cm rainfall, cool growing season with bright sunshine at ripening; Punjab has the highest wheat productivity.
- Uttar Pradesh is the leading producer of wheat.
- Rajasthan leads in bajra and in barley; Maharashtra leads in jowar.

Crop to leading producer state - the stable ones

Crop	Leading producer state
Wheat	Uttar Pradesh
Sugarcane	Uttar Pradesh
Jute	West Bengal
Tea	Assam
Coffee	Karnataka
Rubber	Kerala
Rapeseed-mustard, bajra, barley	Rajasthan
Soybean, gram, total pulses	Madhya Pradesh
Groundnut	Gujarat
Jowar, onion	Maharashtra
Potato, mango	Uttar Pradesh

Crop	Leading producer state
Mulberry silk	Karnataka

TRAP

Rice and cotton ranks move. Cotton is the classic contested one: Gujarat is usually the largest producer, but the largest area under cotton is in Maharashtra, and the two states trade the production lead in bad years. If an option pair asks for cotton, read whether it says area or production.

- India is the world's largest producer of milk, pulses, jute and spices.
- India is the second largest producer of rice, wheat, sugarcane, groundnut, fruits and vegetables.
- China is first in wheat and rice - so 'India is the largest producer of wheat' is always a wrong statement.

Cash crops and plantation crops

Cash crops carry the trade and the history, so they come with dates and nicknames - and RPSC loves both. Sugarcane is the one to be careful with, because it is the only major crop that does not get an MSP.

- Sugarcane - Uttar Pradesh leads in both area and production; sugar recovery is higher in the tropical belt of Maharashtra and Karnataka.
- Sugarcane gets a Fair and Remunerative Price (FRP), not an MSP.
- Jute, the 'golden fibre' - needs 150-200 cm rainfall, grows in the Ganga-Brahmaputra delta.
- India's first jute mill was set up at Rishra, on the Hooghly, in 1855.
- Tea was introduced in Assam in the 1830s; other belts are Darjeeling and the Nilgiris.
- Coffee - Arabica and Robusta - began in the Baba Budan hills of Karnataka.

- Rubber needs over 200 cm rainfall; Kerala dominates.

ASKED IN RAS

Asked in RAS Pre 2024 in the world geography slot - which regions practise Mediterranean agriculture. RPSC clearly tests agricultural typology, not just crop lists. The Indian counterpart of that instinct is the shifting-cultivation names question in the next section.

Soils of India

Soil questions in RAS are always about two things - how the soil formed and what it is rich or poor in. Black soil comes from lava, laterite from leaching, red from crystalline rock. Learn the formation verb for each and half the question answers itself.

Soil to region

Soil	Typical region	Key trait
Alluvial	Northern plains and the deltas	Largest area of any Indian soil
Black (regur)	Deccan plateau - Maharashtra, MP, Gujarat	From basaltic lava; self-ploughing cracks
Red and yellow	Eastern and southern Deccan, Chotanagpur	From crystalline igneous rock; iron oxide colour
Laterite	Malabar coast, Western Ghats, Odisha, north-east	Intense leaching; used as building brick
Arid / desert	Western Rajasthan	Sandy, saline, low organic matter
Forest and mountain	Himalayan slopes	Thin, acidic on higher slopes
Saline and alkaline	Parts of Punjab, Haryana, Gujarat, Rajasthan	Salt crust from over-irrigation and evaporation

Soil	Typical region	Key trait
Peaty and marshy	Kerala backwaters, coastal Odisha, Sundarbans	High organic matter, waterlogged

- Black soil is rich in lime, iron, magnesia and potash - and poor in nitrogen, phosphorus and humus.
- It swells when wet and cracks deeply in summer, which aerates it; hence 'self-ploughing'.
- Laterite is poor in fertility but suits tea, coffee, cashew and rubber.
- Red soils are deficient in nitrogen, phosphorus and humus, like black soil.

TRAP

'Black soil is rich in nitrogen, phosphorus and organic matter' is FALSE and it is the most planted wrong statement in this chapter. Rich in lime-iron-magnesia-potash; poor in N, P and humus. Same for laterite - it is not rich in humus.

YAAD RAKHO

Formation verbs: black = lava cooled, laterite = leached, red = weathered crystalline rock, alluvial = deposited by rivers. Lava, leach, rock, river.

Land use, agro-climatic zones and shifting cultivation

Two sets of numbers get asked here - how the country's land is used, and how many agro-climatic zones there are. The zone count has three different answers depending on who divided the country, so read the question stem carefully.

- Net sown area is about 46 per cent of the reporting area - the single largest land use in India.

- *Cropping intensity = gross cropped area divided by net sown area.*
- *Planning Commission: 15 agro-climatic zones.*
- *ICAR / NARP: 127 agro-climatic zones.*
- *NBSS and LUP: 20 agro-ecological regions.*
- *Agriculture and allied sectors give roughly 18 per cent of gross value added but employ nearly half the workforce.*

Shifting cultivation - the local names

<i>Name</i>	<i>Region</i>
<i>Jhum</i>	<i>North-eastern India</i>
<i>Podu</i>	<i>Andhra Pradesh and Odisha</i>
<i>Bewar and dahiya</i>	<i>Madhya Pradesh</i>
<i>Kumari</i>	<i>Kerala</i>
<i>Walra</i>	<i>Bhils of southern Rajasthan</i>

ASKED IN RAS

Asked in RAS Pre 2013 and again in 2018 - the slash-and-burn cultivation of the Bhils of Rajasthan. Answer: Walra. Because it is the Rajasthan entry in a national list, RPSC returns to it. Learn all five names, not just walra.

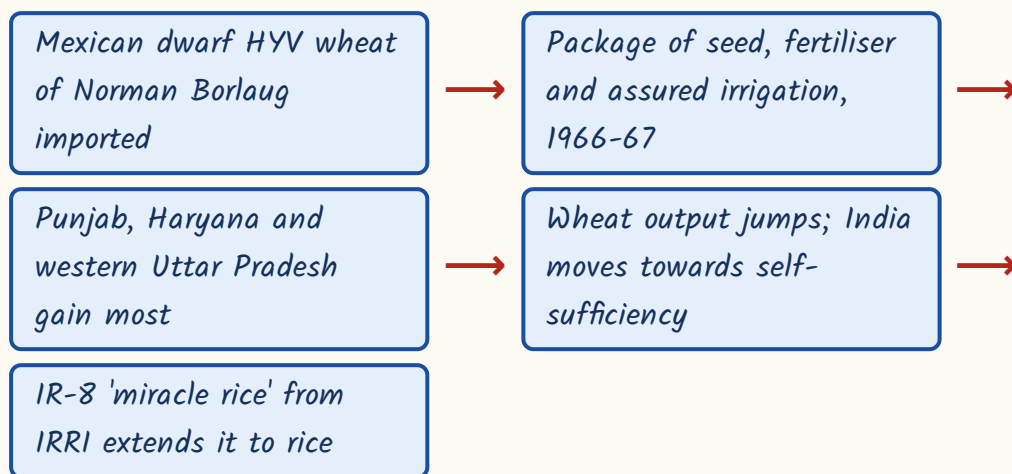
The revolutions

Every colour revolution is simply a product plus a programme plus a person. The Green Revolution is the one with the fullest story, so it gets asked in detail; the rest are asked as one-word matching. Do not overload yourself - one product per colour is enough.

Revolution to product

Revolution	Product	Anchor fact
Green	Foodgrains, especially wheat	1966-67, HYV Mexican dwarf wheat; M. S. Swaminathan
White	Milk	Operation Flood 1970, NDDB, Verghese Kurien, Anand pattern
Blue	Fisheries	Inland and marine fish production
Yellow	Oilseeds	Technology Mission on Oilseeds, 1986
Silver	Eggs and poultry	Poultry expansion
Pink	Prawns and meat processing	Not to be confused with silver
Round	Potato	Short and easy to forget
Grey	Fertilisers	Input side, not output

How the Green Revolution actually ran



- M. S. Swaminathan is called the father of India's Green Revolution.
- Norman Borlaug bred the dwarf wheat varieties in Mexico.
- IR-8 is a rice variety from the International Rice Research Institute in the Philippines - not a wheat variety.
- Operation Flood, 1970, built on Anand-pattern cooperatives under NDDB.

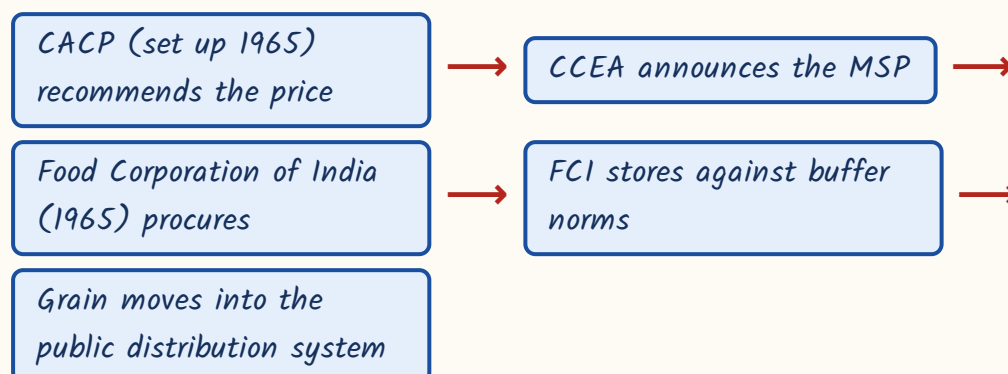
TRAP

The single commonest wrong statement about the Green Revolution: 'it began with the IR-8 variety of wheat'. IR-8 is rice, and Philippine. The wheat came from Mexico.

Prices, marketing and the schemes

The price chain is short and worth memorising as a sequence, because questions attack any one link in it. A body recommends, a cabinet committee announces, an agency procures, and a network distributes.

How a support price reaches a farmer



- MSP is announced for 22 crops, plus the FRP for sugarcane.
- Since 2018-19 MSP is fixed at not less than 1.5 times the A2+FL cost.
- A2+FL = paid-out costs plus the imputed value of family labour.
- e-NAM, the electronic National Agriculture Market, was launched on 14 April 2016.
- e-NAM is run by the Small Farmers' Agribusiness Consortium (SFAC) and links APMC mandis online.

Scheme to year

Scheme	Year	Identifier
Kisan Credit Card	1998	Short-term credit to farmers

Scheme	Year	Identifier
National Food Security Mission	2007-08	Rice, wheat, pulses, coarse cereals
Rashtriya Krishi Vikas Yojana	2007	State-led agriculture planning
Soil Health Card	19 February 2015	Launched from Suratgarh, Rajasthan
PMFBY	Kharif 2016	Replaced NAIS and MNAIS
e-NAM	14 April 2016	Run by SFAC
Operation Greens	2018	Tomato, onion, potato - TOP
PM-KISAN	Announced 24 Feb 2019, effective 1 Dec 2018	Rs 6,000 a year

- PMFBY farmer premium: 2 per cent for kharif, 1.5 per cent for rabi food and oilseed crops, 5 per cent for commercial and horticultural crops.
- PMFBY was made voluntary for loanee farmers in 2020.
- PM-KISAN pays Rs 6,000 a year in three equal instalments of Rs 2,000.
- Paramparagat Krishi Vikas Yojana promotes organic farming; Sikkim became India's first fully organic state in 2016.
- The National Agriculture Policy, 2000 set a 4 per cent annual growth target for the sector.

YAAD RAKHO

Soil Health Card was launched from Suratgarh in Rajasthan. Any national scheme with a Rajasthan launch site is a gift to an RPSC paper-setter - remember the place, not just the date.

Crop research institutes

Institute	Headquarters
National Rice Research Institute (formerly CRR)	Cuttack
Indian Institute of Sugarcane Research	Lucknow

Institute	Headquarters
Central Potato Research Institute	Shimla
Indian Institute of Pulses Research	Kanpur
Indian Institute of Wheat and Barley Research	Karnal
National Research Centre on Seed Spices	Ajmer
National Research Centre on Camel	Bikaner
Central Institute for Arid Horticulture	Bikaner

ASKED IN RAS

Asked in RAS Pre 2016, 2018 and 2023 - the national research centres located in Rajasthan. Seed Spices at Ajmer, Camel at Bikaner, Arid Horticulture at Bikaner. Three papers, the same three institutes. Learn the city with each.

CURRENT LINK

Current link, as of September 2026: PM-KISAN crossed its 22nd instalment, released in March 2026 from Guwahati, and a further instalment followed in mid-2026. The instalment number changes every four months, so never learn it - learn Rs 6,000 a year in three instalments of Rs 2,000, which does not change.

REVISION RECAP

1. Kharif is sown with the monsoon (June-July) and harvested September-October; rabi is sown October-December and harvested April-June; zaid fills the gap.
2. Kharif = rice, jowar, bajra, maize, cotton, jute, groundnut, tur, soybean. Rabi = wheat, barley, gram, mustard, peas, linseed.
3. Rice needs over 100 cm rain and about 25 degrees C; West Bengal grows aus, aman and boro; Punjab has the highest rice and wheat yields.
4. Stable leaders: wheat and sugarcane UP, jute West Bengal, tea Assam, coffee Karnataka, rubber Kerala, groundnut Gujarat, soybean and pulses MP, mustard/bajra/barley Rajasthan.

5. India is the world's largest producer of milk, pulses, jute and spices; second in rice, wheat, sugarcane and groundnut.
6. Sugarcane gets FRP, not MSP; UP leads in area and production, Maharashtra and Karnataka in recovery.
7. First jute mill at Rishra, 1855; first cotton textile mill at Mumbai, 1854 - keep the pair straight.
8. Alluvial soil covers the largest area; black soil comes from Deccan lava and is poor in nitrogen, phosphorus and humus; laterite is formed by intense leaching.
9. Planning Commission 15 agro-climatic zones, ICAR/NARP 127, NBSS and LUP 20 agro-ecological regions; net sown area about 46 per cent.
10. Shifting cultivation: jhum north-east, podu AP-Odisha, bewar/dahiya MP, kumari Kerala, walra among the Bhils of Rajasthan.
11. Green Revolution 1966-67 with Mexican dwarf wheat; Swaminathan the father in India, Borlaug the breeder; IR-8 is rice from IRRI.
12. White = milk (Operation Flood 1970, Kurien, NDDB), Yellow = oilseeds (Technology Mission 1986), Blue = fish, Silver = eggs, Pink = prawns, Round = potato, Grey = fertilisers.
13. CACP (1965) recommends MSP for 22 crops; since 2018-19 it is at least 1.5 times A2+FL; FCI (1965) procures and stores.
14. e-NAM launched 14 April 2016 under SFAC; PMFBY from kharif 2016 with 2 / 1.5 / 5 per cent farmer premiums.
15. PM-KISAN - Rs 6,000 a year in three instalments of Rs 2,000; announced 24 February 2019, effective from 1 December 2018.
16. Soil Health Card launched 19 February 2015 from Suratgarh, Rajasthan; Sikkim India's first fully organic state, 2016.
17. Rajasthan's national research centres: Seed Spices Ajmer, Camel Bikaner, Arid Horticulture Bikaner - asked in 2016, 2018 and 2023.